

# **CS689 Final Project Report**

Student: Susan Jia

Course: MET CS689 Designing and Implementing a Data Warehouse

Project Title: Amazon Order & Revenue Data Warehouse

## **1. Project Overview**

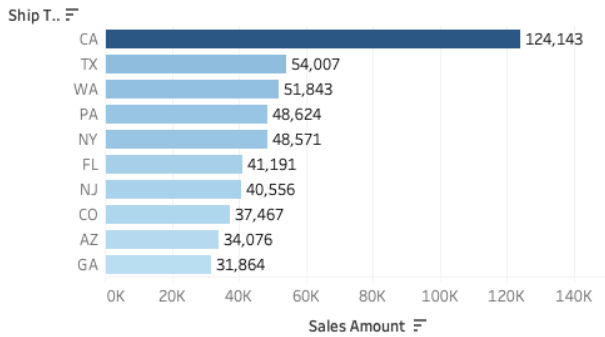
The purpose of this project was to design and implement a dimensional data warehouse using Amazon order and delivery data. The warehouse was built to support business analytics, KPI reporting, SQL querying, and Tableau dashboard visualization.

This project demonstrates the complete data warehouse lifecycle, including raw data ingestion, staging, data cleaning, dimensional modeling, fact table loading, business intelligence reporting, and final executive dashboard creation.

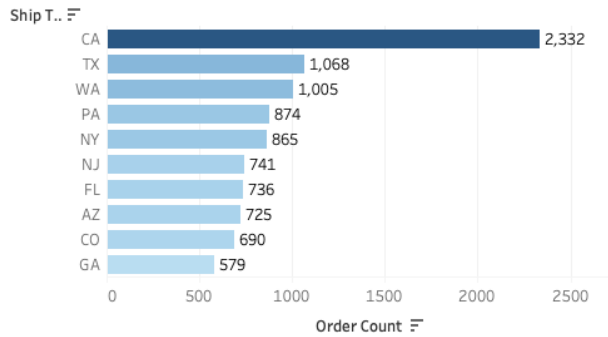
The final solution transforms operational transaction data into a structured warehouse environment that supports business decision-making.

## Amazon Order & Revenue Performance Dashboard

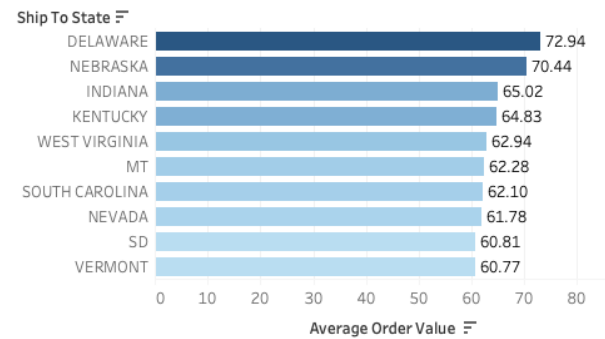
### Top Destination States by Sales



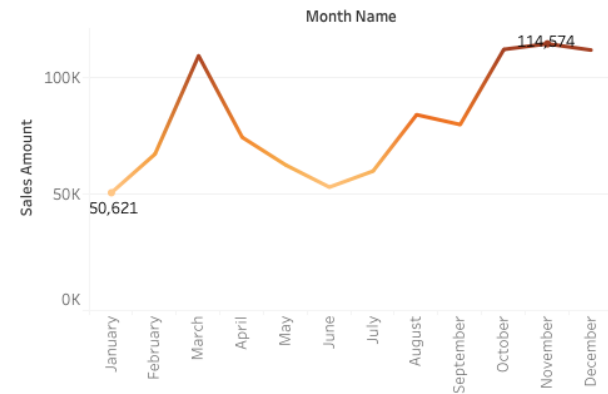
### Order Volume by State



### Average Order Value by State



### Monthly Revenue Trend



## 2. Data Sources

Two source CSV files were used for this project.

amazon\_orders.csv

This file contains Amazon order transaction data, including:

order\_id

order\_date

product information

quantity

display price

sales amount

tax amount

fulfillment method

shipping destination

amazon\_delivery.csv

This file contains delivery operation data, including:

agent age

agent rating

weather

traffic

vehicle type

delivery area

delivery time

delivery category

These two files were integrated to support both sales analytics and delivery performance analysis.

### **3. Business Questions**

The warehouse was designed to answer the following business questions:

1. Which states generate the highest total sales revenue?
2. Which states have the highest order volume?
3. Which states have the highest average order value?
4. How does revenue change over time?
5. How do traffic and weather affect delivery performance?

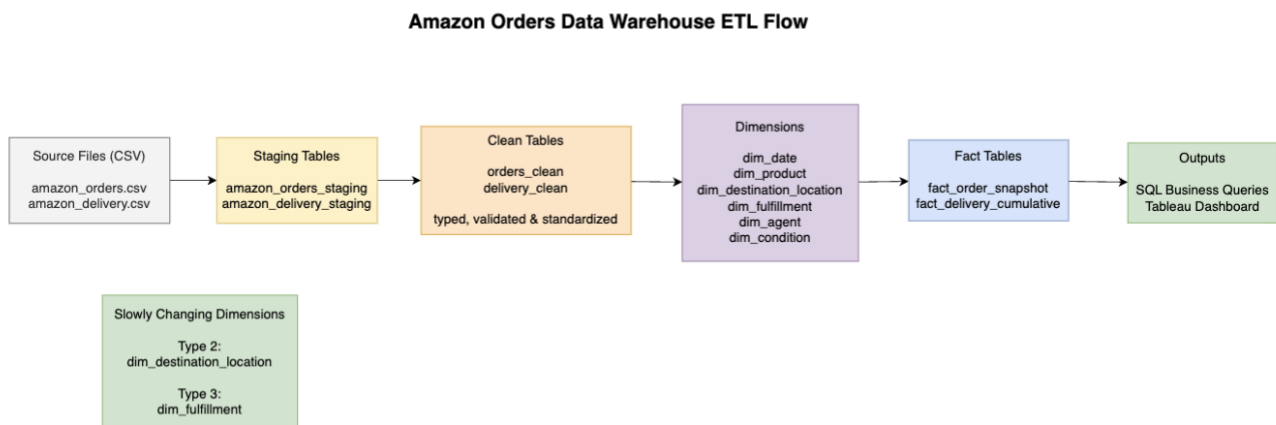
These business questions guided the dimensional model, SQL design, and dashboard development.

## 4. Data Warehouse Architecture

The final warehouse uses a dimensional star-schema design with fact tables and dimension tables.

Warehouse Flow:

Source Files → Staging Tables → Clean Tables → Dimensions → Fact Tables → SQL Analysis → Tableau Dashboard



This layered design improves data quality, maintainability, and analytical performance.

## 5. Fact Tables

Two fact tables were created.

fact\_order\_snapshot

Type: Transactional Snapshot Fact Table

Grain: One row per order transaction

Measures:

Quantity

display\_price

sales\_amount

order\_count

tax\_amount

This table supports revenue and sales reporting.

fact\_delivery\_cumulative

Type: Cumulative Fact Table

Grain: One row per delivery process record

Measures:

delivery\_time

agent\_rating

This table supports delivery efficiency and operational analysis.

## **6. Dimension Tables**

The warehouse contains six dimension tables.

dim\_date

Stores calendar attributes:

full\_date

month\_num

month\_name

quarter\_num

year\_num

weekday\_name

dim\_product

Stores product identifiers:

Asin

sku

dim\_destination\_location

Stores shipping destination attributes:

ship\_to\_country

ship\_to\_state

ship\_to\_city

postal\_code

SCD Type 2 implemented.

dim\_fulfillment

Stores fulfillment methods.

SCD Type 3 implemented.

dim\_agent

Stores delivery agent classifications:

agent\_age\_group

agent\_rating\_band

dim\_condition

Stores delivery environment factors:

Weather

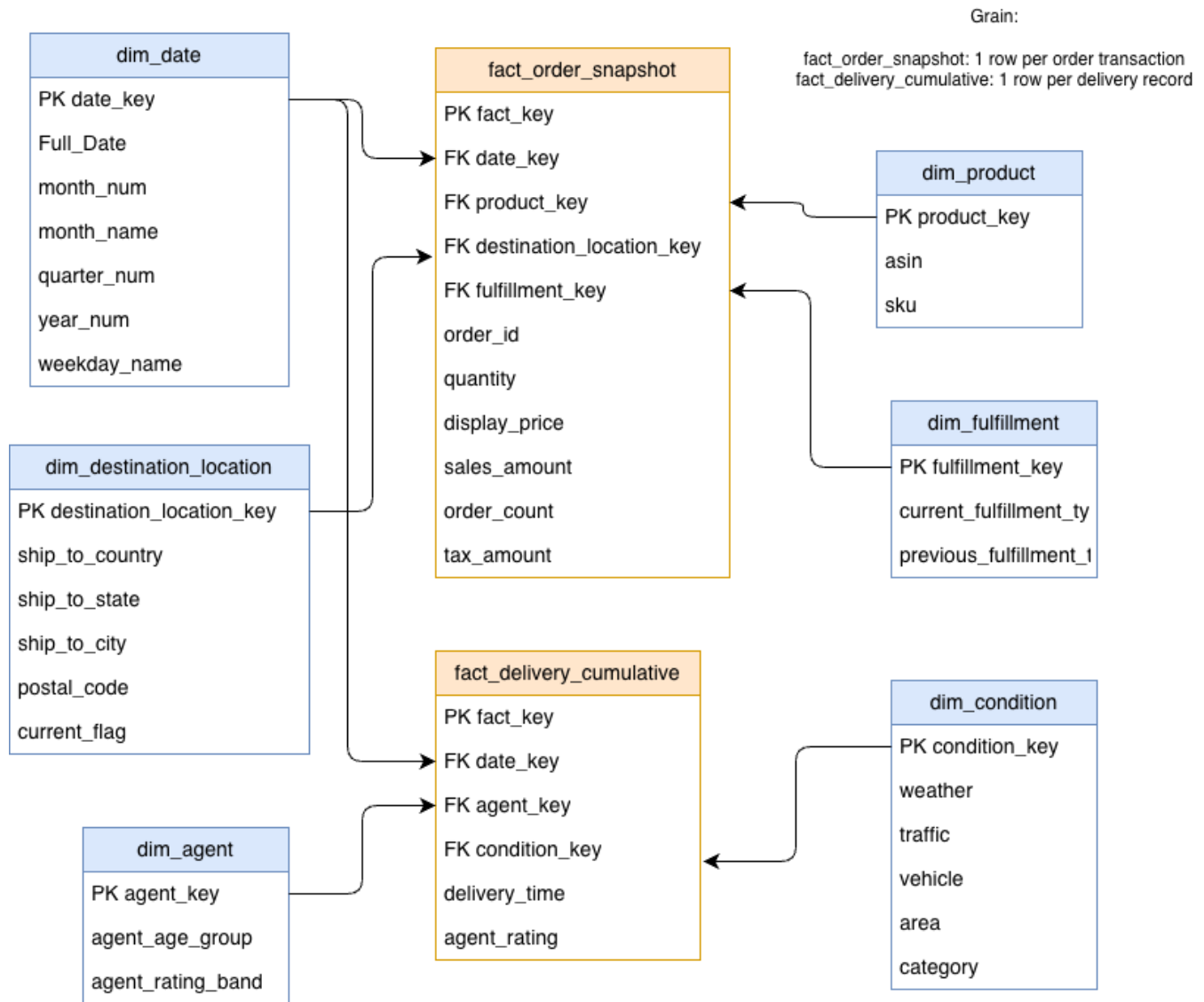
Traffic

Vehicle

Area

Category

# Amazon Sales Warehouse Dimensional Model



SCD Type 2: dim\_destination\_location

SCD Type 3: dim\_fulfillment



## **7. ETL Process**

The ETL pipeline was implemented in SQL and followed these steps.

### **Step 1 – Load Raw Data**

CSV files were imported into staging tables:

amazon\_orders\_staging

amazon\_delivery\_staging

### **Step 2 – Clean and Standardize Data**

Data cleaning included:

trim spaces

convert dates

convert numeric values

replace null values

standardize state names

calculate sales metrics

Clean tables created:

orders\_clean

delivery\_clean

### **Step 3 – Load Dimension Tables**

Dimension tables were loaded first using distinct values and surrogate keys.

## Step 4 – Load Fact Tables

Fact tables were loaded by joining clean tables to dimension keys.

This ETL structure ensures data consistency and warehouse performance.

## **8. Slowly Changing Dimensions**

Two Slowly Changing Dimension methods were implemented.

SCD Type 2 – dim\_destination\_location

Used to preserve historical destination changes.

When location values changed:

old record was expired

end\_date updated

current\_flag changed to N

new active row inserted

This preserves history over time.

| Data Output Messages Notifications |                                         |                         |                       |                      |                             |                    |                  |                               |  |
|------------------------------------|-----------------------------------------|-------------------------|-----------------------|----------------------|-----------------------------|--------------------|------------------|-------------------------------|--|
| Showing rows: 1 to 1000            |                                         |                         |                       |                      |                             |                    |                  |                               |  |
|                                    | destination_location_key<br>[PK] bigint | ship_to_country<br>text | ship_to_state<br>text | ship_to_city<br>text | ship_to_postal_code<br>text | start_date<br>date | end_date<br>date | current_flag<br>character (1) |  |
| 1                                  | 6713                                    | US                      | AK                    | Anchorage            | 99503-5774                  | 2026-04-20         | [null]           | Y                             |  |
| 2                                  | 7562                                    | US                      | AK                    | Anchorage            | 99504-2317                  | 2026-04-20         | [null]           | Y                             |  |
| 3                                  | 4302                                    | US                      | AK                    | Anchorage            | 99508-5021                  | 2026-04-20         | [null]           | Y                             |  |
| 4                                  | 7152                                    | US                      | AK                    | Anchorage            | 99508                       | 2026-04-20         | [null]           | Y                             |  |
| 5                                  | 2324                                    | US                      | AK                    | Anchorage            | 99504-3405                  | 2026-04-20         | [null]           | Y                             |  |
| 6                                  | 1422                                    | US                      | AK                    | Delta Junction       | 99737-0881                  | 2026-04-20         | [null]           | Y                             |  |
| 7                                  | 3902                                    | US                      | AK                    | Eagle River          | 99577-8504                  | 2026-04-20         | [null]           | Y                             |  |
| 8                                  | 9040                                    | US                      | AK                    | Eagle River          | 99577-7798                  | 2026-04-20         | [null]           | Y                             |  |
| 9                                  | 9853                                    | US                      | AK                    | Juneau               | 99801-2015                  | 2026-04-20         | [null]           | Y                             |  |
| 10                                 | 3194                                    | US                      | AK                    | Kenai                | 99611-8417                  | 2026-04-20         | [null]           | Y                             |  |
| 11                                 | 5815                                    | US                      | AK                    | Palmer               | 99645-7700                  | 2026-04-20         | [null]           | Y                             |  |
| 12                                 | 10145                                   | US                      | AK                    | Palmer               | 99645-8940                  | 2026-04-20         | [null]           | Y                             |  |
| 13                                 | 8579                                    | US                      | AK                    | Wasilla              | 99687                       | 2026-04-20         | [null]           | Y                             |  |
| 14                                 | 4772                                    | US                      | AK                    | Wasilla              | 99623-4916                  | 2026-04-20         | [null]           | Y                             |  |
| 15                                 | 5481                                    | US                      | AK                    | Wasilla              | 99623-0767                  | 2026-04-20         | [null]           | Y                             |  |
| 16                                 | 4564                                    | US                      | AK                    | Wasilla              | 99654                       | 2026-04-20         | [null]           | Y                             |  |
| 17                                 | 7634                                    | US                      | AL                    | Albertville          | 35951-7348                  | 2026-04-20         | [null]           | Y                             |  |
| 18                                 | 7773                                    | US                      | AL                    | Anniston             | 36207-5008                  | 2026-04-20         | [null]           | Y                             |  |
| 19                                 | 1393                                    | US                      | AL                    | Anniston             | 36207-1184                  | 2026-04-20         | [null]           | Y                             |  |
| 20                                 | 5388                                    | US                      | AL                    | Athens               | 35613-6630                  | 2026-04-20         | [null]           | Y                             |  |
| 21                                 | 6022                                    | US                      | AL                    | Auburn               | 36832-8668                  | 2026-04-20         | [null]           | Y                             |  |
| 22                                 | 8386                                    | US                      | AL                    | Auburn               | 36830-6831                  | 2026-04-20         | [null]           | Y                             |  |
| 23                                 | 9484                                    | US                      | AL                    | Auburn               | 36830-0335                  | 2026-04-20         | [null]           | Y                             |  |
| 24                                 | 8984                                    | US                      | AL                    | Auburn               | 36830-0244                  | 2026-04-20         | [null]           | Y                             |  |
| 25                                 | 1947                                    | US                      | AL                    | Bay Minette          | 36507                       | 2026-04-20         | [null]           | Y                             |  |
| 26                                 | 5229                                    | US                      | AL                    | Birmingham           | 35242-3450                  | 2026-04-20         | [null]           | Y                             |  |
| 27                                 | 1193                                    | US                      | AL                    | Birmingham           | 35244-2285                  | 2026-04-20         | [null]           | Y                             |  |
| 28                                 | 8327                                    | US                      | AL                    | Birmingham           | 35244-1600                  | 2026-04-20         | [null]           | Y                             |  |
| 29                                 | 311                                     | US                      | AL                    | Birmingham           | 35206-3954                  | 2026-04-20         | [null]           | Y                             |  |
| 30                                 | 6809                                    | US                      | AL                    | Birmingham           | 35242-4402                  | 2026-04-20         | [null]           | Y                             |  |
| 31                                 | 3340                                    | US                      | AL                    | Birmingham           | 35233-2705                  | 2026-04-20         | [null]           | Y                             |  |
| 32                                 | 3483                                    | US                      | AL                    | Clanton              | 35046-4612                  | 2026-04-20         | [null]           | Y                             |  |
| 33                                 | 5280                                    | US                      | AL                    | Columbiana           | 35051-3940                  | 2026-04-20         | [null]           | Y                             |  |
| 34                                 | 1635                                    | US                      | AL                    | Cullman              | 35055-6237                  | 2026-04-20         | [null]           | Y                             |  |

## SCD Type 3 – dim\_fulfillment

Used to track previous and current fulfillment values in the same row.

Columns:

current\_fulfillment\_type

previous\_fulfillment\_type

This allows recent change tracking without inserting new rows.

| Data Output Messages Notifications |                                |                                  |                                   |
|------------------------------------|--------------------------------|----------------------------------|-----------------------------------|
| ≡+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL          |                                |                                  |                                   |
|                                    | fulfillment_key<br>[PK] bigint | current_fulfillment_type<br>text | previous_fulfillment_type<br>text |
| 1                                  | 1                              | AFN_UPDATED                      | AFN                               |
| 2                                  | 2                              | MFN_UPDATED                      | MFN                               |

## 9. SQL Analytics Results

### Business Question 1

Which states generate the highest revenue?

California generated the highest total revenue, followed by Texas and Washington.

### Business Question 2

Which states have the highest order volume?

California and Texas recorded the highest order counts.

### Business Question 3

Which states have the highest average order value?

Several smaller states showed strong average order value despite lower volume.

### Business Question 4

How does revenue change over time?

Revenue peaked in Q4, especially November, showing strong seasonal demand.

## Business Question 5

How do traffic and weather affect delivery performance?

Heavy traffic and poor weather were associated with longer delivery times.

## **10. Tableau Dashboard**

A final Tableau dashboard was developed using warehouse output tables.

Dashboard Features:

Top states by revenue

Top states by order count

Average order value by state

Monthly revenue trend

Filtering

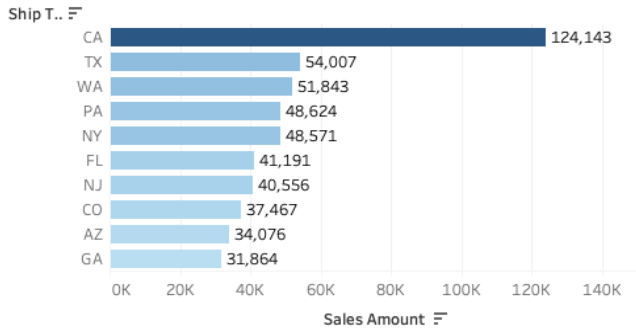
Hierarchy drill-down

Calculated fields

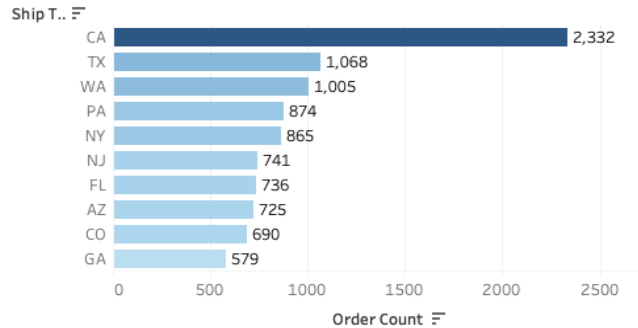
Multiple chart types

## Amazon Order & Revenue Performance Dashboard

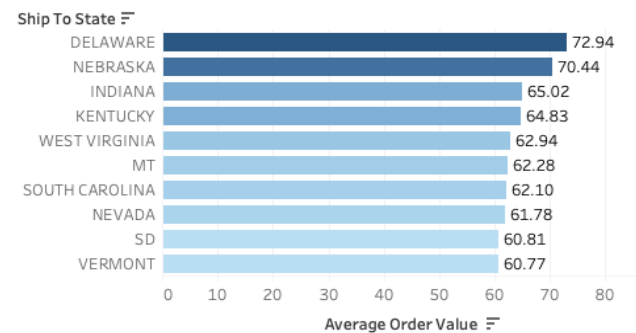
### Top Destination States by Sales



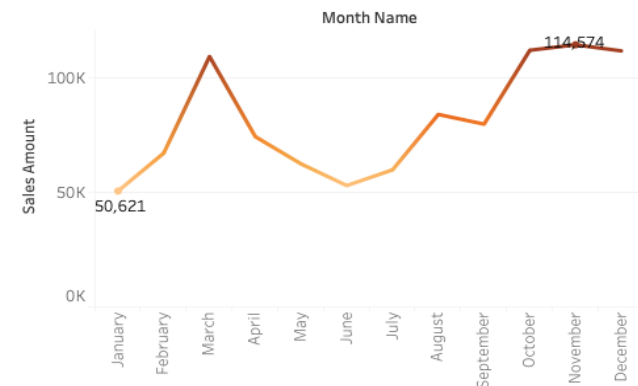
### Order Volume by State



### Average Order Value by State



### Monthly Revenue Trend



The dashboard improves usability and supports executive reporting.

## 11. Key Business Recommendations

1. Increase inventory and marketing focus in top revenue states.
2. Expand premium product campaigns in high-value states.
3. Prepare inventory and staffing before Q4 demand peaks.
4. Use traffic and weather forecasts for smarter delivery scheduling.
5. Continue growing mid-tier states with strong upside potential.

## **12. Final Reflection**

This project significantly improved my understanding of dimensional modeling, ETL development, SQL analytics, and business intelligence reporting.

I learned how raw operational data can be transformed into structured fact and dimension tables that support business decision-making.

I also gained practical experience implementing Slowly Changing Dimensions, designing Tableau dashboards, and translating business questions into measurable KPIs.

Overall, this project helped connect technical warehouse design with real-world analytics strategy and decision support.